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Topic: **Wazuh-SIEM-Deployment-Endpoint-Monitoring-and-File-Integrity-Analysis**

Course Code: CIP-A-107

Course Title: Security Operation Center

Instructor: Aminu Idris

1. Introduction

The aim of this lab was to set up Wazuh and understand how it can be used for security monitoring in a Security Operations Centre (SOC). I used Ubuntu and Windows 11 as virtual machines in VirtualBox. Ubuntu was used as the Wazuh server, while Windows 11 was used as the endpoint to be monitored. The lab was also meant to help me understand how File Integrity Monitoring (FIM) works and how it can be used to detect changes made to files.

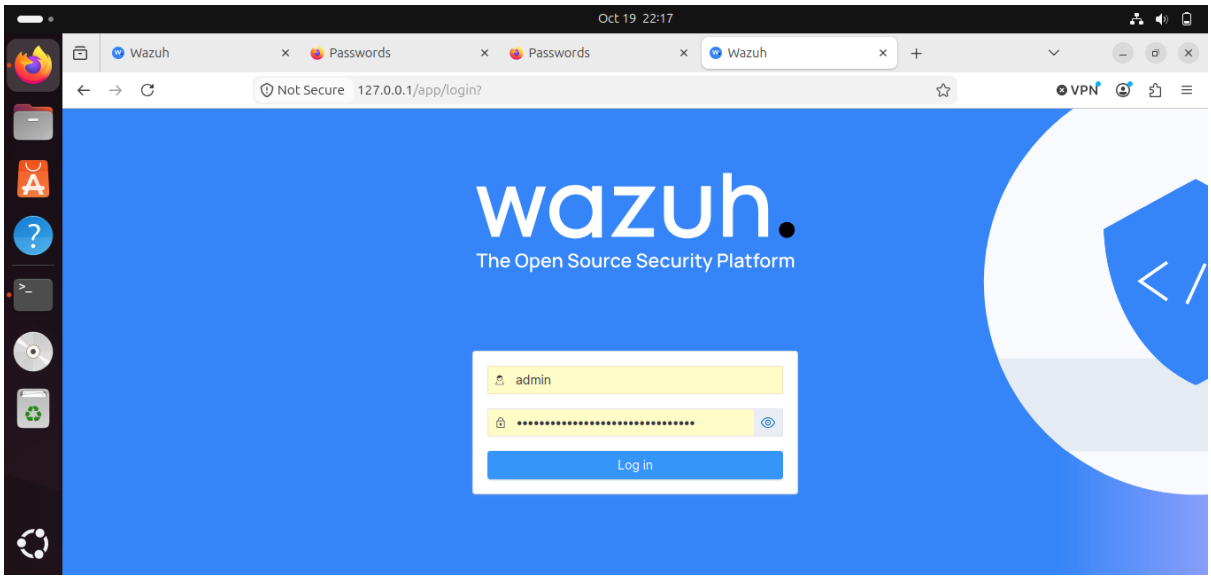
2. Procedures

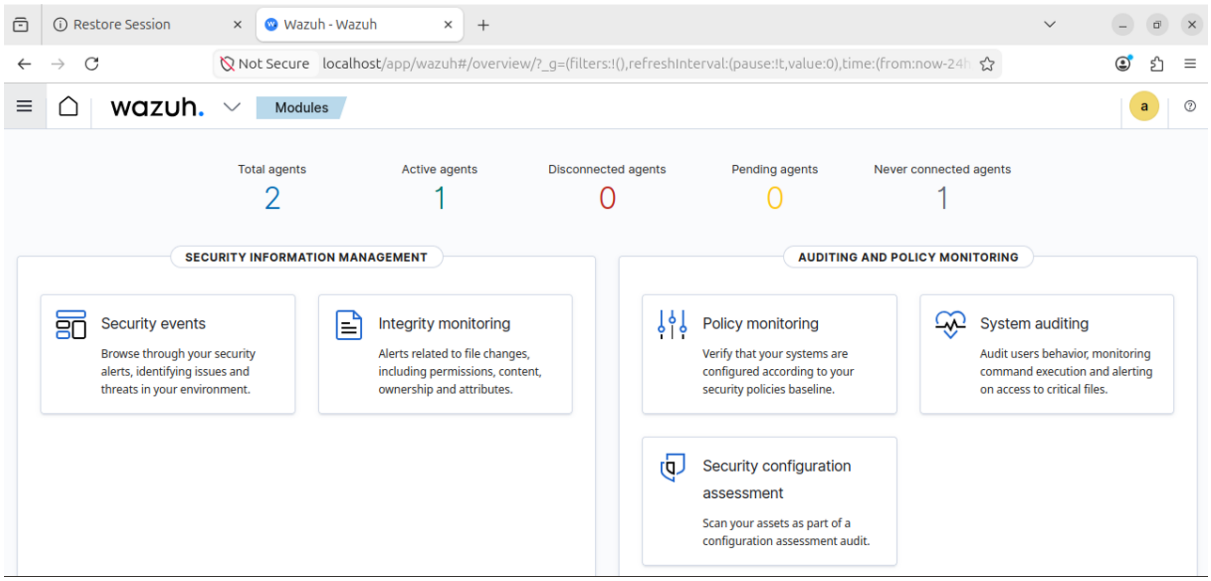
Step 1: Setting Up the Wazuh Server

I used my Ubuntu virtual machine in VirtualBox to install and set up Wazuh. After the installation, I checked the Wazuh Manager service to confirm that it was running. The service showed as active, which indicated that the manager had started successfully.

Step 2: Accessing the Wazuh Dashboard

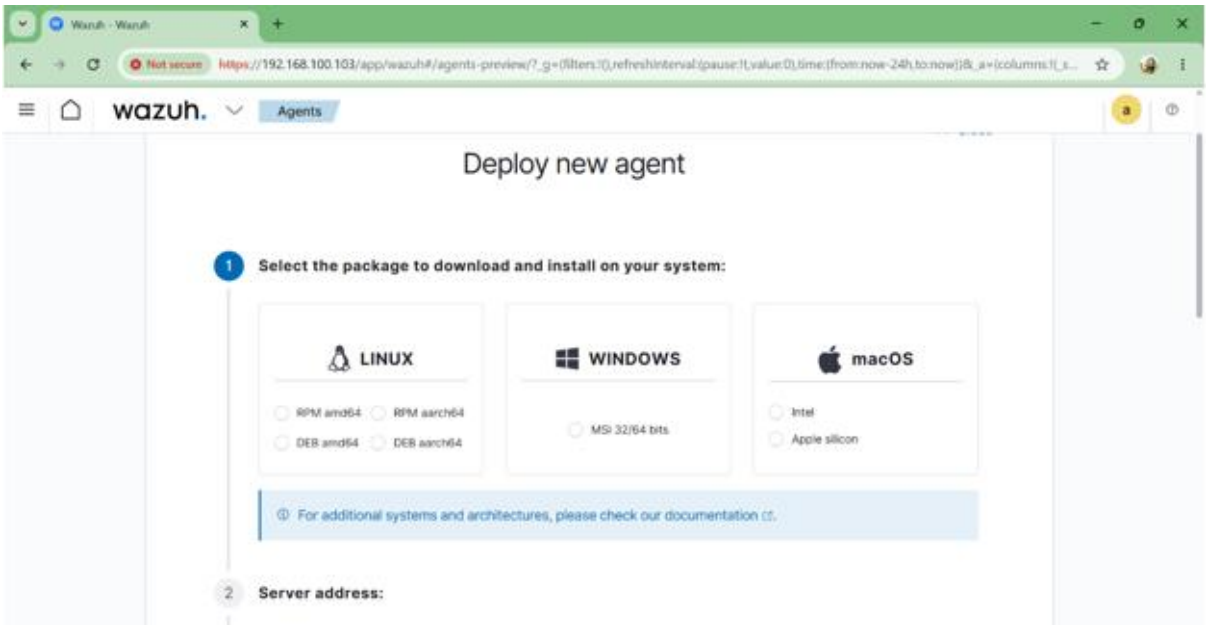
After setting up the Wazuh server, I tried to access the Wazuh dashboard through a web browser using the server IP address. This allowed me to reach the dashboard interface and check the Wazuh environment.





Step 3: Setting Up the Windows 11 Virtual Machine

I used a Windows 11 virtual machine in VirtualBox as the endpoint for the lab. I installed the Wazuh Agent on the Windows 11 virtual machine so that it could communicate with the Wazuh server running on Ubuntu.



3. Challenges Encountered

One of the challenges I faced was viewing the expected file monitoring events on the Wazuh dashboard. Although I was able to set up the Wazuh server and get the Windows 11 virtual machine to appear on the dashboard, I could not confirm that the expected FIM alerts were showing.

Because of this, I could not fully verify the file modification events through the dashboard. I would need to troubleshoot the FIM configuration and event collection to understand why the alerts were not appearing. This was the main part of the lab that I could not fully verify.

Lab Challenge & Analysis Questions

1. Alert Triage

When Wazuh FIM detects a file modification, the alert can contain details that help an analyst understand what changed. These details include the file path, file hashes such as SHA1, SHA256 or MD5, file size before and after the change, and the modification time. The event field can also show that the file was modified. These details help an analyst investigate the change and determine whether it is suspicious. In my lab, I could not confirm the expected file modification alert on the Wazuh dashboard, so I was unable to inspect these details in an actual alert.

2. False Positive Identification

Two folders that can generate many FIM alerts are C:\Windows\Temp and C:\Program Files.

The first one is C:\Windows\Temp. This folder contains temporary files created by Windows and applications. These files can be created and deleted regularly, which can make Wazuh generate many alerts that are not actual threats.

The second one is C:\Program Files. This folder contains installed applications. When software is updated or installed, some files in the folder may change. These changes can also generate alerts even when they are part of normal computer activities.

3. Proactive Defense

I can use Wazuh FIM to monitor important folders such as Documents, Desktop, and NetworkShares. This can help me notice when many files are modified or renamed within a short time.

For example, ransomware may change the extensions of files from .docx to .locked after encrypting them. Wazuh FIM can help detect these file changes if the folders are being monitored. I can then check the alerts to find out whether the changes are suspicious. This can help in identifying possible ransomware activity before more damage is done.

4. Tool Comparison

Windows Event Logs store information about activities happening on a Windows computer, while Wazuh provides a central dashboard where security information from connected computers can be monitored.

With Windows Event Logs, I may need to check the logs on individual computers unless they have been configured for central collection. Wazuh makes it easier to view alerts from connected endpoints in one place. It can also use rules to generate alerts and help connect file changes with other security events, such as login or process activities.

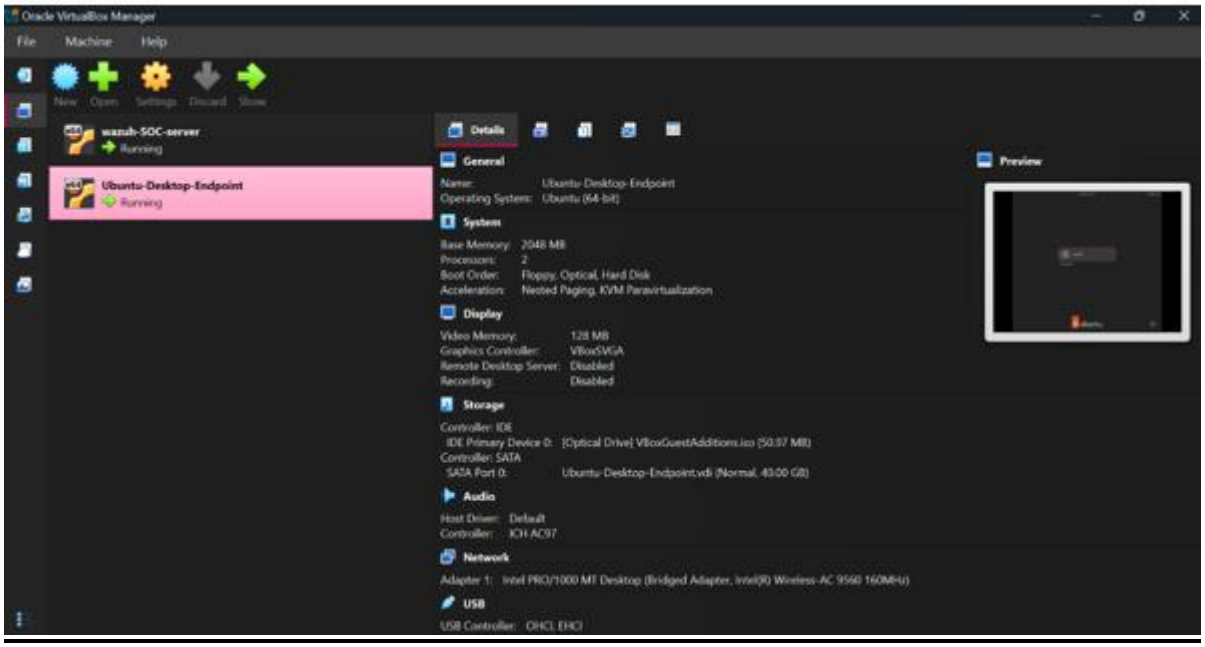
Another advantage of Wazuh is that it provides searchable security information and can be configured to perform automated responses. This makes it useful for monitoring and investigating security events in a SOC.

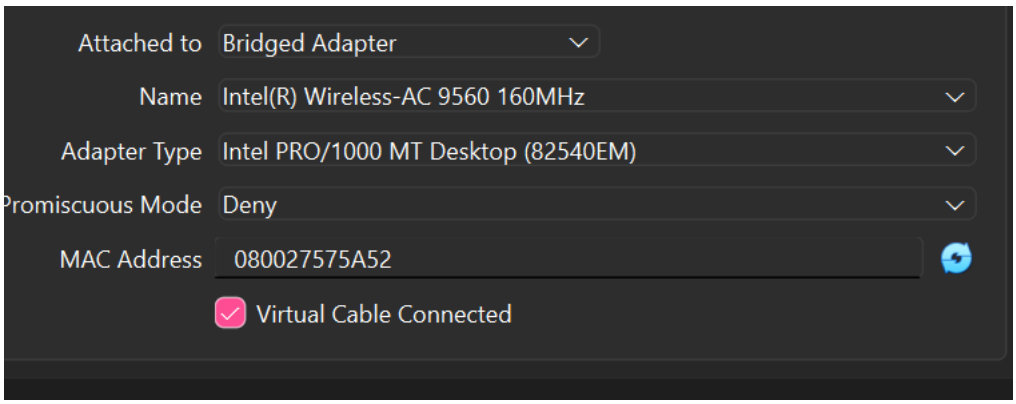
Conclusion

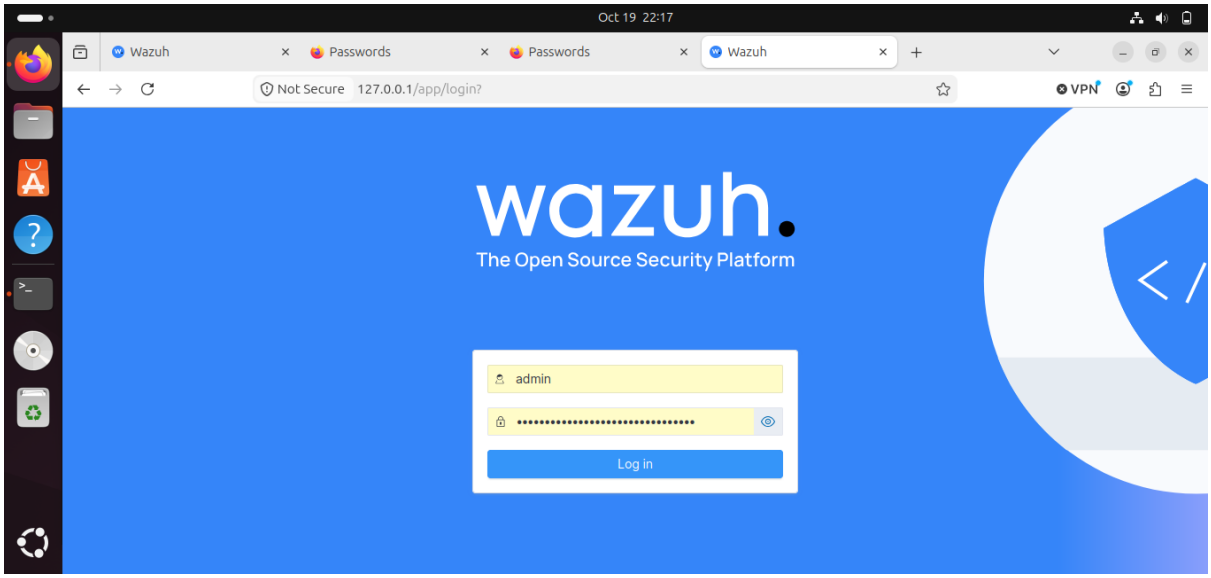
From this lab, I learned how Wazuh can be used to monitor a computer and detect changes made to files. I also learned that some folders can generate many alerts because of normal activities, so it is important to choose the folders to monitor carefully. I understood how FIM

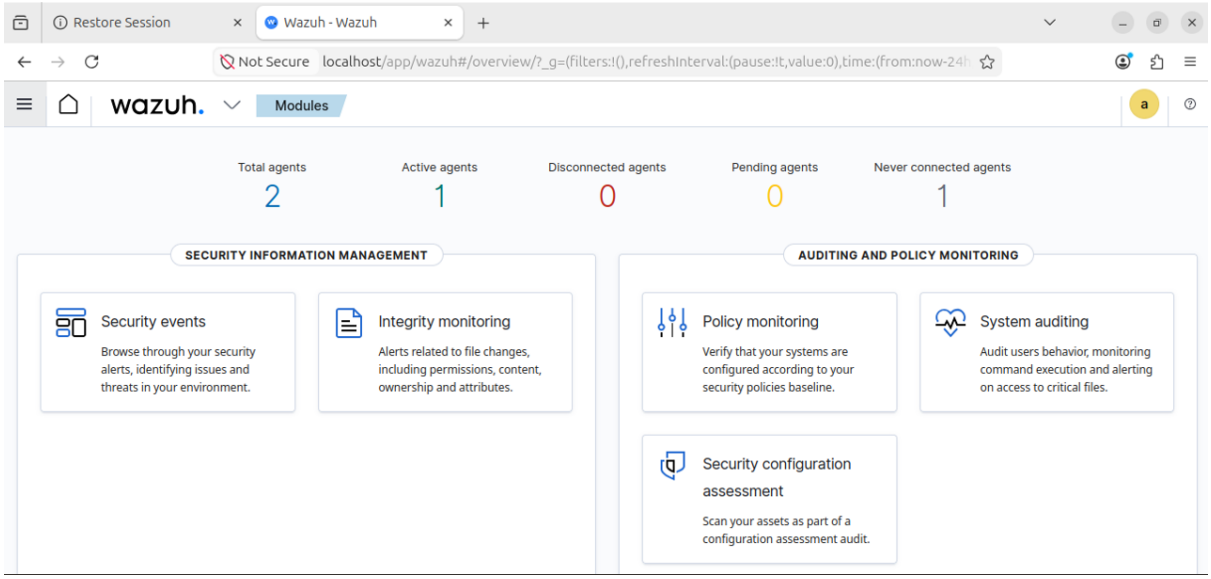
can help identify possible ransomware activity by detecting unusual file changes. Although I was able to set up the Wazuh environment and connect my Windows 11 virtual machine, I had challenges viewing the expected FIM alerts on the dashboard. I still need to troubleshoot this part to understand why the alerts were not showing. Overall, the lab helped me understand the basic use of Wazuh for security monitoring and the importance of checking alerts during an investigation.

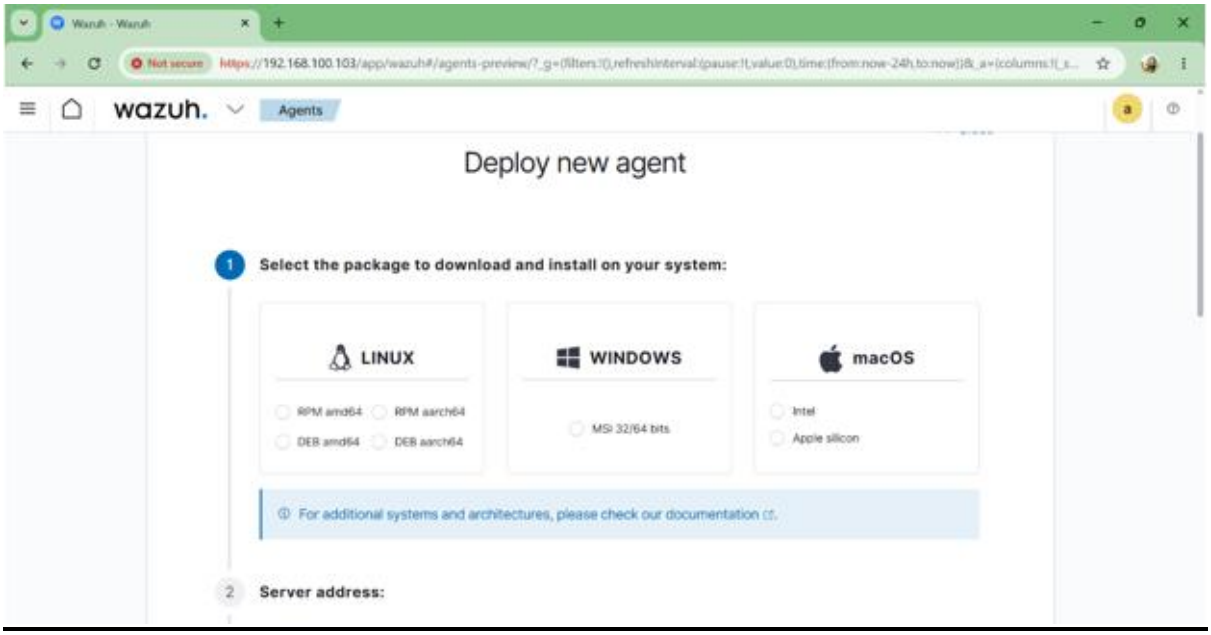
Screenshots

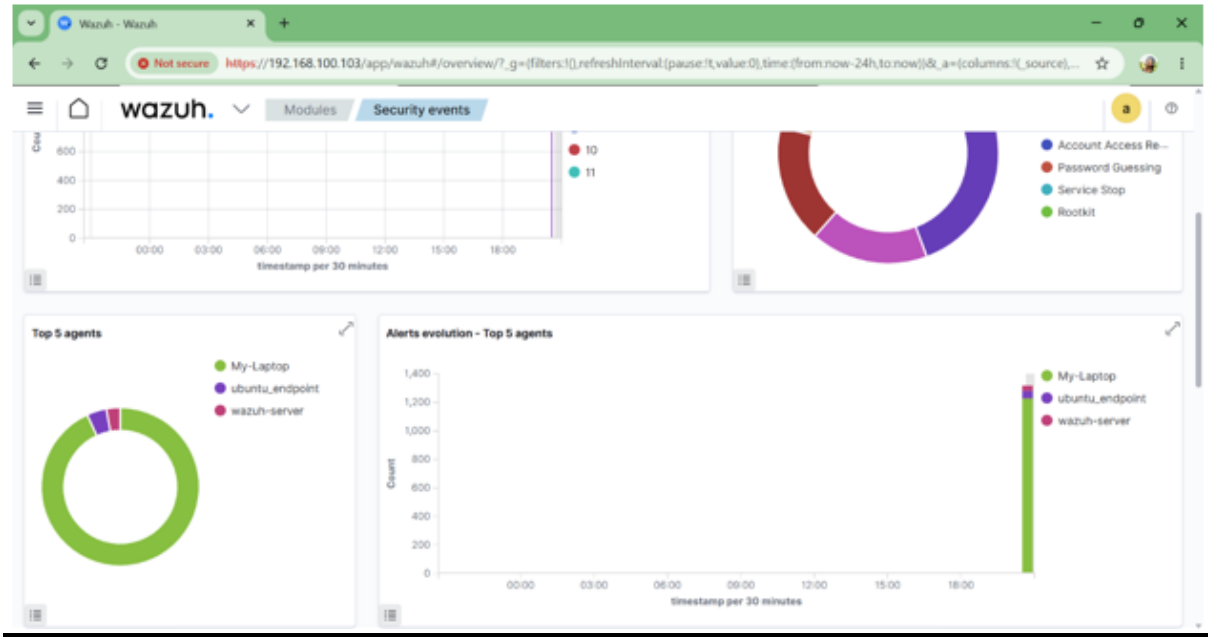




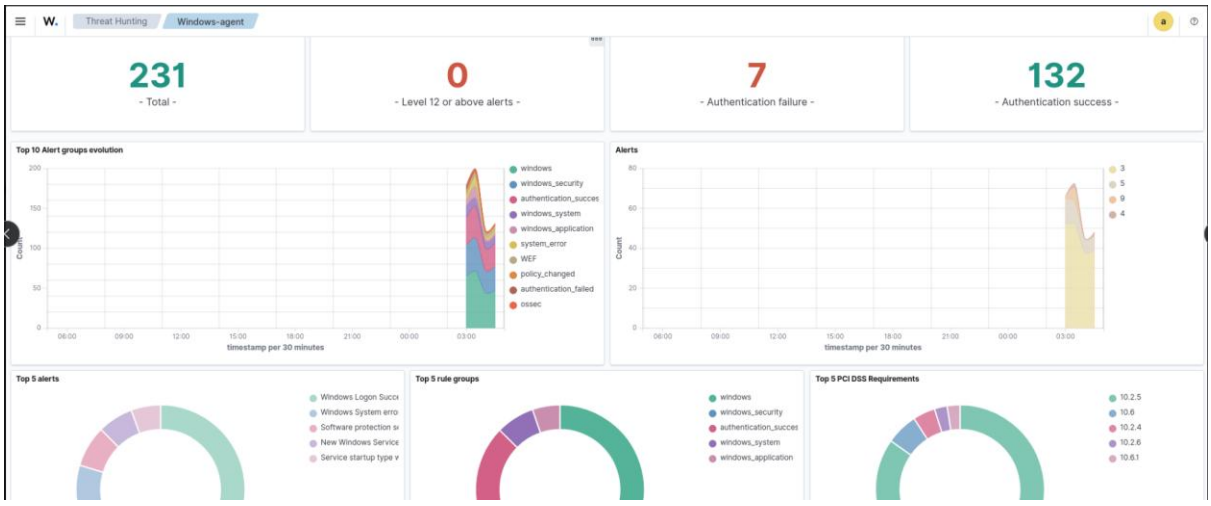

Oct 19 22:17
Wazuh
Passwords
Passwords
Wazuh
Not Secure
127.0.0.1/app/login?
VPN
wazuh.
The Open Source Security Platform
admin
Log in


Restore Session
Wazuh - Wazuh
Not Secure
localhost/app/wazuh#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h
wazuh.
Modules
Total agents
2
Active agents
1
Disconnected agents
0
Pending agents
0
Never connected agents
1
SECURITY INFORMATION MANAGEMENT
Security events
Browse through your security alerts, identifying issues and threats in your environment.
Integrity monitoring
Alerts related to file changes, including permissions, content, ownership and attributes.
AUDITING AND POLICY MONITORING
Policy monitoring
Verify that your systems are configured according to your security policies baseline.
System auditing
Audit users behavior, monitoring command execution and alerting on access to critical files.
Security configuration assessment
Scan your assets as part of a configuration assessment audit.


wazuh.
Agents
Deploy new agent
1 Select the package to download and install on your system:
LINUX
RPM amd64
RPM aarch64
DEB amd64
DEB aarch64
WINDOWS
MSI 32/64 bits
macOS
Intel
Apple silicon
For additional systems and architectures, please check our documentation.
2 Server address:


wazuh.
Modules
Security events
timestamp per 30 minutes
Account Access Re...
Password Guessing
Service Stop
Rootkit
Top 5 agents
My-Laptop
ubuntu_endpoint
wazuh-server
Alerts evolution - Top 5 agents
Count
timestamp per 30 minutes


Wazuh - Wazuh
Not secure
https://192.168.100.103/app/wazuh#/overview/?tab=general&_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))&tabView=...
wazuh.
Modules
Security events
My-Laptop
wazuh-server
ubuntu_endpoint
My-Laptop
wazuh-server
Count
timestamp per 30 minutes


Threat Hunting
Windows-agent
231
- Total -
0
- Level 12 or above alerts -
7
- Authentication failure -
132
- Authentication success -
Top 10 Alert groups evolution
windows
windows_security
authentication_succes
windows_system
windows_application
system_error
WEF
policy_changed
authentication_failed
ossec
timestamp per 30 minutes
Alerts
Count
timestamp per 30 minutes
Top 5 alerts
Top 5 rule groups
Top 5 PCI DSS Requirements
10.2.5
10.6
10.2.4
10.2.6
10.6.1